

Matthew Oj

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SKILLS

Languages: Python, SQL, JavaScript/TypeScript, PHP, HTML/CSS

Frameworks & Cloud: FastAPI, Next.js, React, Astro, Laravel, Docker, GitHub Actions, AWS, GCP, Vercel, Netlify, CI/CD

Databases & Data Tools: PostgreSQL, SQLite, MongoDB, BigQuery, Snowflake, dbt, Apache Airflow, Snowpipe, DuckDB, XGBoost, SHAP, Vertex AI Generative Models

EXPERIENCE

Senior Data Engineer (Backend Infrastructure)

Apr 2026 – present

VAFO Digital

London, UK

- Led migration of fragmented backend orchestration (**Snowflake** procedures, **PHP** services, serverless functions) to a unified self-hosted platform on **AWS EC2** using **Docker**, provisioning isolated staging/production environments with an **Aurora RDS** metadata store and **IAM**-scoped access, consolidating scheduling for **100+** services
- Architected a reporting service (**FastAPI**, **AWS Lambda**, **Mangum**) automating scheduled delivery across **10** client accounts, eliminating manual ad hoc requests
- Built a shared **Python** library packaged for reuse, standardising error handling and logging across **30+** services and **4** integrations (**Klaviyo**, **DynamoDB**, **SFTP**, **S3**)
- Developed a self-hosted **dbt** documentation portal, generating manifest artefacts via dbt docs, publishing to **S3**, and serving through **AWS Amplify** with a custom **HTML**, **CSS** and **JavaScript** front end, giving stakeholders searchable, always-current data lineage

Software Engineer, Data

Apr 2025 – Apr 2026

VAFO Digital

London, UK

- Collaborated with the Data Science Lead to productionise **4** pipelines (Marketing Mix Modelling, personalisation, churn) using **dbt** feature engineering, **XGBoost**/**SHAP** training, and **Snowflake** Model Registry for versioning
- Built and hosted **3** client-facing analytics platforms (**Python**, **Streamlit**) on Snowflake, with high-performance querying against event tables exceeding **200M+** rows to deliver real-time insights to non-technical stakeholders
- Developed **Python AWS Lambda** services syncing backend data to external systems (**HubSpot**, **Klaviyo**), supporting **250K+** users

Software Engineer | maisonprive.london

Apr 2026 – June 2026

Maison Privé Concierge Limited (Freelance)

Remote

- Built and deployed a luxury concierge marketing site (**Astro**, **TypeScript**) on **Vercel**, with **SSR API** routes handling enquiries across **9** services in **3** categories
- Engineered a dynamic enquiry form with per-service field schemas, validation, rate limiting, and spam filtering, blocking **95%+** of bot submissions
- Built a dual-sink pipeline routing enquiry to **Mailgun** and **Airtable** in real time

Data Engineer

Nov 2023 – Apr 2025

EssenceMediacom

London, UK

- Built a **Cloud Run**-hosted web app for table schema generation, cutting pipeline build time by **30%** via no-code tool integration
- Improved **Apache Airflow** DAG deployment speed by **40%** through **CI/CD** pipeline optimization with **GitHub Actions**
- Developed a **REST API**-driven alert system with **Cloud Run** to notify teams of Adverity DataStream failures, boosting incident response time
- Developed a secure web application for the Data Strategy team, enabling **OAuth**-based access for users across **5+** agencies

IT Support Analyst

Jan 2021 – Oct 2023

University of Greenwich

London, UK

- Developed static **HTML/CSS** webpages to display IT support documentation and helpline scripts, reducing resolution time for frequent issues
- Improved security compliance by **75%** through multi-factor authentication resets using **Microsoft Azure AD** (now **Microsoft Entra ID**)

PROJECTS

Finance Tracker | FastAPI, dbt, PostgreSQL, Next.js, Docker | [GitHub](#)

April 2026

- Designed a production-style data stack ingesting raw financial data through **dbt** models into aggregated **PostgreSQL** marts, served via a **JWT**-secured **FastAPI** and visualised in a **Next.js** dashboard, all containerised with **Docker Compose**
- Wrote a **pytest** suite covering auth, token validation, all endpoints, and pagination against a fully mocked database - 13/13 tests passing.

Sims 4 Mod Manager | Python, SHA-256, DBPF Parsing | [GitHub](#)

April 2026

- Built a desktop mod manager scanning **88GB+** of mods with **SHA-256** caching, conflict detection via **DBPF** package header parsing, and smart load-order assignment

ACTIVITIES

WPP & Google Cloud Hackathon (3rd Place) | Python, Cloud Run, Vertex AI, Vertex AI Generative Models, Streamlit

Sep 2024

- Designed and deployed a **Cloud Run** service pulling real-time data from public **BigQuery** tables to dynamically generate prompts for **Vertex AI**, rendering AI-generated advertising creatives through a **Streamlit** frontend, placed **3rd** of all competing teams

Global Game Jam Hackathon - ML & Game Developer | Python, OpenCV, C#, Unity

Jan 2024

- Built a **Python/OpenCV** ML script using camera input to detect laughs, aligned with the hackathon's humour theme
- Contributed **Unity/C#** game objects and animations to the game prototype

Engineers Without Borders - Project Lead Engineer | MATLAB, SolidWorks

Sep 2020 – May 2021

- Engineered a **MATLAB** script to optimize thermal efficiency in material selection – increasing solar cooker design efficiency by **20%**.
- Led a team of **5** engineers to design a solar cooker prototype with phase-changing materials using **SolidWorks**

EDUCATION

University of Greenwich

Sep 2020 – May 2023

Bachelor of Engineering, Civil Engineering, 2.1 Upper Second-Class Honours

London, UK

Dissertation (First Class): Assessing the vulnerability of catchments during flash flood events in South England using Machine Learning methods